
Seasonal Patterns Affect Road Fatalities Differently in Germany and the United Kingdom

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Abstract

This study examines whether seasonal fluctuations in fatal road accident risk differ between Germany and the United Kingdom. Using 2024 national data, accident counts were normalized by population and vehicle kilometers traveled to ensure comparability. A hierarchical Poisson model was employed to account for regional heterogeneity while focusing on relative seasonal variation rather than absolute fatality levels. By isolating seasonal risk from national baselines, this research provides a comparative framework for understanding how national contexts mediate the relationship between seasonal factors and road safety. Results indicate that Germany's fatal accident rate exhibits significantly higher seasonal sensitivity than the United Kingdom's, suggesting that national contexts differentially mediate the impact of seasonal factors on road safety.

1. Introduction

Road traffic fatalities impose a severe global socioeconomic burden and are projected to remain a leading cause of mortality (World Health Organization (WHO), 2018). Seasonal factors such as weather, daylight and traffic volume are widely assumed to influence the risk of fatal road accidents. While numerous studies examine seasonal patterns within individual countries (Shafay et al., 2024; Copsey et al., 2025), systematic cross-country comparisons remain limited. Understanding these differences can provide insights relevant to traffic safety policy and risk management.

This study compares seasonal risk variations in Germany and the United Kingdom (UK) using 2024 government datasets. To facilitate a robust comparison, we partitioned administrative units into synthetic regions of approximately 500,000 residents, balancing urban density against rural

sparsity. Accident counts are normalized by both population size and traffic exposure, measured as vehicles per billion kilometer of road, to account for differences in exposure between regions and countries. Regional heterogeneity and uneven accident counts are addressed using a hierarchical Poisson modeling framework. This allows estimation of mean fatal accident rates and relative changes in risk across seasons. The analysis emphasizes seasonal variation in risk rather than absolute fatality levels.

The remainder of the paper presents the data sources, preprocessing steps, and modeling methodology in Section 2, and provides a comparative analysis of seasonal effects on fatal road accidents in Germany and UK in Section 3. All analysis code and derived datasets are publicly available in the project's [GitHub repository](#).

2. Data and Methods

This section outlines our data preprocessing, the normalization techniques applied, and the hierarchical Poisson framework used to assess seasonal fatality differentials.

2.1. Data Sources

We utilize official 2024 government records for comprehensive national coverage. German data are collected from the Unfallatlas (Federal Statistical Office of Germany, 2024a), comprising 268,519 personal injury collisions characterized by location, time and the conditions at the scene of the accident. UK data are gathered from the Department for Transport's STATS19 database (UK Department for Transport, 2024a), comprising 100,927 personal injury collisions in 2024. The dataset integrates three relational components: collision circumstances (location, time, conditions), involved vehicles, and casualty details. Regional boundaries and population statistics are obtained from the Federal Agency for Cartography and Geodesy and the Federal Statistical Office for Germany (Federal Agency for Cartography and Geodesy, 2024; Federal Statistical Office of Germany, 2024b), and the Office for National Statistics for the UK (Office for National Statistics, UK, 2024b;a).

Road network and traffic exposure information, including road length, road type, and vehicle count data, are obtained

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from OpenStreetMap (OSM) extracted by Geofabrik (Geofabrik, 2024), the Federal Highway Research Institute (BAST) (Federal Highway Research Institute (BAST), 2024), and UK Department for Transport traffic statistics (Department for Transport, UK, 2024).

2.2. Data Preprocessing

To ensure cross-national comparability, UK collision data was reduced to German reporting standards. Vehicle types were aggregated into binary indicators to identify the involvement of specific categories per incident.

Area classification for the German dataset was generated according to the official UK methodology (UK Department for Transport, 2024b), flagging accidents as urban if located within municipalities exceeding 10,000 inhabitants. This was achieved by spatially joining incident coordinates with municipal geometries (Federal Agency for Cartography and Geodesy, 2024; Federal Statistical Office of Germany, 2024a) and population data. Road types were reconciled by extracting motorway geometries from OSM data (Geofabrik, 2024), German accidents within a 50-meter distance were classified as motorway incidents, while all others were designated as federal highways or secondary roads.

The final processed dataset focuses exclusively on fatal collisions, organized by meteorological seasons to provide a consistent basis for comparative analysis.

2.3. Normalization Procedures

To compare regional fatal accident risks, counts are normalized by resident population and vehicle traffic to assess both societal risk and infrastructure safety quality (International Transport Forum (ITF), 2023).

To mitigate the Modifiable Areal Unit Problem (MAUP) (Fotheringham & Wong, 1991) and stabilize variance, original regional counts $A_i, i \in \{1, \dots, N\}$ are redistributed into population-balanced units $A_j, j \in \{1, \dots, M\}$ of approximately 500,000 inhabitants by splitting or merging directly adjacent regions appropriately. This homogenization ensures a consistent statistical signal and prevents low-population outliers from skewing the national average. The population-normalized rate per 100,000 inhabitants is:

$$R_{\text{pop},j} = \frac{A_j}{P_j} \times 10^5, \quad (1)$$

where A_j and P_j are the unit-specific accident counts and population. For national c population normalization per month m we use $(\sum_j A_{c[j],m}) \cdot (\sum_j P_{c[j],m})^{-1}$.

Traffic normalization accounts for varying vehicle activity on the Strategic Road Network (SRN) (Highways England, 2017) in the UK and the Bundesfernstraßen (Federal Trunk Roads) (Fernstraßen-Bundesamt, 2025) in Germany, includ-

ing OSM motorway, trunk, and primary classes. This selection ensures high-fidelity Average Annual Daily Flow (AADF) data across consistent infrastructure standards. The length-weighted AADF for unit j is:

$$\text{AADF}_j = \frac{\sum_{r \in \mathcal{R}_j} \text{AADF}_r \cdot L_r}{\sum_{r \in \mathcal{R}_j} L_r}, \quad (2)$$

where $\mathcal{R}_j$ denotes the set of major road segments in re-districted region j , AADF_r is the segment-specific flow, and L_r is the segment length.

To estimate exposure, we calculate the monthly Vehicle Kilometers Traveled (VKT) for each unit j :

$$\text{VKT}_{j,m} = \text{AADF}_j \cdot L_j \cdot D_m, \quad (3)$$

where $L_j = \sum_{r \in \mathcal{R}_j} L_r$ is the total length of the major road network within unit j , D_m is the number of days in month m . This ensures that the subsequent modeling accounts for the sheer volume of opportunities for accidents to occur.

This framework enables robust cross-national comparison by minimizing exposure uncertainty. Excluding minor roads with intermittent flow data shifts the focus toward motorized traffic risk on primary streets rather than non-motorized dynamics (International Transport Forum (ITF), 2023).

2.4. Statistical Modeling

Modeling accident counts via a Poisson distribution is a standard approach for independent rare events occurring over a defined exposure (McCullagh & Nelder, 1989; Lord et al., 2005). We model monthly fatal accident counts $Y_{j,m}$ using a hierarchical Poisson regression with $\text{VKT}_{j,m}$ as the exposure offset. To align with international benchmarks expressed per billion vehicle kilometers (BVK), we define the offset as $E_{j,m} = \text{VKT}_{j,m} \cdot 10^{-9}$:

$$Y_{j,m} \sim \text{Poisson}(E_{j,m} \cdot \lambda_{j,m}), \quad (4)$$

where the log-rate $\log \lambda_{j,m}$ is decomposed into:

$$\log \lambda_{j,m} = \mu_{c[j]} + \sigma_{c[j]} \eta_j + \gamma_{c[j],m} + \tau \tilde{\delta}_{j,m}. \quad (5)$$

The country baseline $\mu_{c[j]}$ uses a weakly informative $\mathcal{N}(1, 2)$ prior, centering expectation at $\exp(1) \approx 2.7$ fatalities per BVK (International Transport Forum (ITF), 2023). Spatial heterogeneity $\eta_j \sim \mathcal{N}(0, 1)$ is scaled by a country-specific $\sigma_{c[j]}$. To isolate relative risk fluctuations, the country-level seasonal effects $\gamma_{c[j],m}$ and regional deviations $\tilde{\delta}_{j,m}$ are sum-to-zero constrained annually. This forces these parameters to represent purely monthly deviations, where a log-rate of 0 represents the annual baseline. Consequently, the Relative Risk (RR) discussed in Sections 3.2 and 3.3 is defined as $\exp(\gamma_{c[j],m})$, where a value of 1.0 represents the annual average risk. Spatial scale parameter $\sigma_{c[j]}$ and the regional seasonal scale parameter τ

are assigned Half-Normal priors to enforce strictly positive support. Specifically, $\tau \sim \text{Half-Normal}(0.5)$ provides shrinkage for the regional effects, preventing the model from over-fitting noise in units with sparse accident data.

Posterior inference was performed using the No-U-Turn Sampler (NUTS) in **NumPyro**. We ran 4 independent chains with 2,000 warmup and 4,000 sampling iterations each, totaling 16,000 posterior draws.

2.5. Analytical Framework

Building on this framework, we first compare normalization strategies to demonstrate the implications of using AADF-based normalization in the Poisson model versus population-based normalization for hourly and area-type analyses. This distinction avoids bias from applying aggregate daily flows to intra-day traffic peaks and mitigates sensor-coverage limitations on minor urban roads. We then present comparative seasonal risk profiles, followed by vehicle-specific seasonal patterns, temporal distributions of fatalities, road-type seasonal sensitivity, and seasonal patterns by area type.

3. Results

While Germany recorded a higher absolute count of road fatalities in 2024 (2,458) than the UK (1,502), the following sections evaluate these figures through the normalized risk and seasonal decomposition established in our methodology.

3.1. Compare Normalizations

An initial comparison of seasonal accident rates (Table 1) demonstrates that the observed national trends are robust to the choice of normalization technique. Germany exhibits a pronounced seasonal gradient, characterized by a summer peak and a winter minimum, whereas the UK maintains a comparatively stable risk profile throughout the year. These divergent patterns persist across all applied metrics, prompting a more granular analysis of monthly posterior effects.

country	norm.	winter	spring	summer	autumn	all
GER	pop.	0.6	0.72	0.92	0.74	2.98
UK		0.53	0.52	0.6	0.52	2.16
GER	AADF	2.39	2.91	3.41	3.20	2.98
UK		1.66	1.69	1.85	1.72	1.73

Table 1. Seasonal fatal accident rates per 100,000 citizens and fatalities per BVK, by normalization technique (Section 2.3).

3.2. Comparative Seasonal Risk Profiles

Estimated seasonal effects $\exp(\gamma_{c[j],m})$ reveal higher volatility in Germany than the UK (Figure 1). On this scale, 1.0 denotes the annual baseline risk. German relative risk fluctuates from a January minimum of ≈ 0.8 to a July peak of

≈ 1.3 , while the UK remains between ≈ 0.9 and 1.2. The 95% Highest Density Intervals (HDI) for Germany widen during the third-quarter peak a bit, indicating increased posterior uncertainty. In contrast, the UK maintains a bit narrower intervals throughout the annual cycle, suggesting a more stable seasonal risk profile.

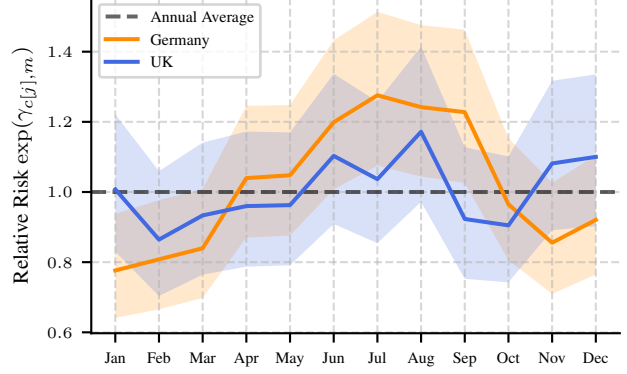


Figure 1. Seasonal posterior mean relative risk $\exp(\gamma_{c[j],m})$ for DE and UK, shaded areas denote 95% HDI, both showing risk relative to a 1.0 annual average.

3.3. Vehicle-Specific Seasonal Patterns

Decomposing the aggregate trends via our traffic-adjusted Poisson model reveals that Germany’s higher seasonal volatility is driven by distinct vehicle-specific dynamics (Figure 2). Most notably, German motorcycle risk exhibits a high-amplitude cycle with a sharp summer peak, whereas the UK profile remains comparatively flat. Passenger car risk follows a similar but attenuated pattern: Germany displays a clear late-summer elevation, while UK rates remain consistent with the annual baseline throughout the year.

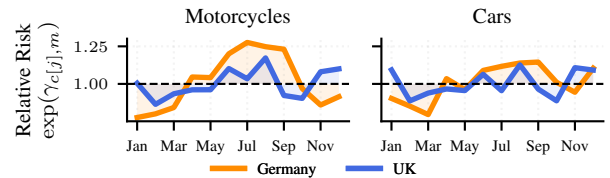


Figure 2. Posterior mean relative risk $\exp(\gamma_{c[j],m})$ for motorcycles and cars relative to the annual baseline (1.0).

3.4. Temporal Distribution of Fatalities

To examine how fatal accident risk varies over the course of the day, we analyze population-normalized hourly fatality rates (Figure 3). Both countries exhibit a dual-peak structure. However, the seasonal dynamics differ: While the UK retains a consistent profile shape and magnitude across all seasons, Germany exhibits a distinct radial expansion in summer, with rates consistently exceeding the UK’s during daylight hours, followed by a contraction in winter where the distributions largely overlap.

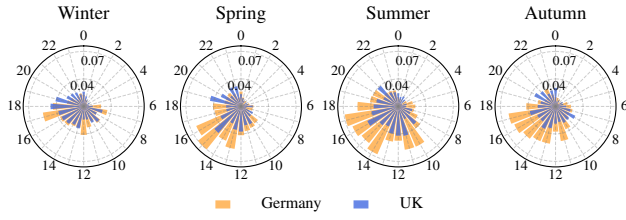


Figure 3. Hourly fatal accident rates by season. Radial axis represents mean hourly accident rate per 100,000 inhabitants.

3.5. Road Type Seasonal Sensitivity

Seasonal sensitivity differs strongly by road type (Figure 4). While motorways account for a minority of total fatalities, the Germany/UK rate ratio for motorway accidents peaks sharply in summer. In contrast, other road types maintain relatively stable cross-country ratios throughout the year.

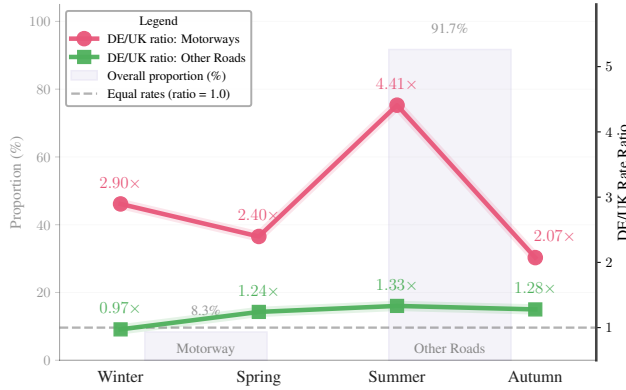


Figure 4. Dual-axis chart comparing seasonal fatal accident road rate ratios between Germany and the UK, categorized by road type.

3.6. Seasonal Patterns by Area Type

To further decompose seasonal effects, we analyze fatal accidents separately for urban and rural areas, normalized by national population (Figure 5). Germany shows strong seasonality in both settlement types, with a clear summer peak. In the UK, rural areas follow the same pattern, while urban areas show the opposite ordering, with highest rates in winter and lowest in summer. Rural rates exceed urban rates in the UK across all seasons, whereas in Germany urban rates are consistently higher. Overall, seasonal variation is much stronger in Germany than in the UK. Across all seasons and both countries, cars dominate, while bicycles and motorcycles make up a larger share in summer.

4. Discussion & Conclusion

Using a hierarchical Poisson model with exposure adjustment, this study finds that seasonal variation in fatal road accident risk is substantially stronger in Germany than in the UK. While both countries exhibit elevated risk during summer months, the estimated seasonal effects in Germany show a markedly higher amplitude, whereas the UK displays a comparatively stable risk profile across the year.

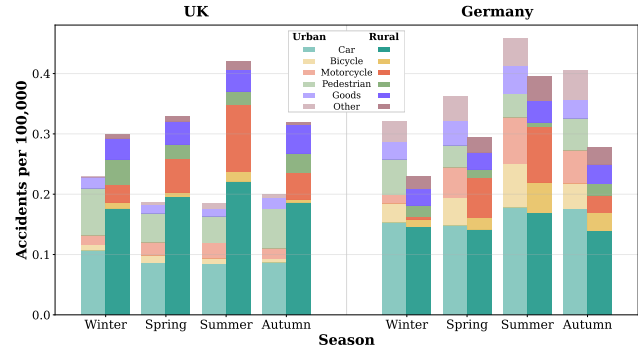


Figure 5. Fatal accident rates per 100,000 inhabitants by season and area type, with bars stacked by vehicle type.

This indicates that seasonality plays a more dominant role in shaping fatal accident risk in Germany.

The supporting analyses are consistent with this finding. Germany exhibits a distinct risk surge across all daylight hours in summer, whereas the UK profile remains more constant. Moreover, seasonal sensitivity is most pronounced on motorways, pointing to high-speed and long-distance travel as key contributors to the observed differences. Additionally, extreme seasonal peaks for motorcycles and moderate elevations for passenger cars indicate strong seasonal sensitivity in German accident dynamics across both urban and rural areas, whereas in the UK motorcycle-related increases are confined to rural contexts. Germany's elevated urban rates, particularly for Pedestrians and Cyclists, suggest higher active-travel density. However, because the "Urban-Rural" distinction is based on geographic built-up area rather than functional road type, these results may reflect settlement structure and traffic density rather than inherent road safety. Accurate risk isolation requires normalization against traffic volume and road classification as current datasets primarily cover motorways and federal roads.

Further limitations include the use of annual AADF data for monthly exposure, capturing spatial traffic intensity but ignoring intra-annual fluctuations in traffic volume, so some observed seasonality may reflect unmeasured exposure rather than per-unit risk. However, as this bias is consistent across both study areas, it likely does not explain the significantly stronger seasonal effect identified in Germany. Additionally, German datasets identified fatal collisions by vehicle type but did not report precise fatality counts or which vehicles incurred fatalities. As a result, the analysis focused on fatal collisions rather than total deaths, limiting vehicle-level detail but still providing a robust basis for cross-national comparison.

Overall, consistent results across analyses indicate that seasonality has a substantially stronger impact on fatal road accident risk in Germany than in the UK, emphasizing the influence of seasonally varying mobility and environmental conditions.

Contribution Statement

Philipp Bauer developed the population and traffic normalization procedures, including the spatial redistribution algorithm and the hierarchical Poisson model. Karl von Barvalle performed data preprocessing and the exploratory data analysis, comparative normalization assessment, and managed the project organization. Jens Kienzle contributed to data preprocessing and analyzed the temporal distribution of fatalities and vehicle-specific seasonal patterns. Katarzyna Nowicka conducted data exploration and analyzed seasonal patterns by road type and temporal fatality distributions. Bilal El Ali executed data analysis focused on area-type seasonal patterns and the seasonal sensitivity of different road types. All authors contributed equally to the final report.

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